

A Reinforcement Learning Approach to Wi-Fi Rate Adaptation Using the REINFORCE Algorithm

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Abstract—In this paper, we propose a novel approach for rate adaptation in 802.11 wireless networks based on the reinforcement learning algorithm, REINFORCE. Our approach leverages a more comprehensive set of observations, including received signal strength, contention window size, current MCS and throughput, enabling a nuanced response to varying network conditions and hence leads to optimal network throughput. The unique application of REINFORCE allows our methodology to iteratively learn optimal actions under different states, resulting in enhanced rate adaptation in dynamically changing environments. We utilize ns-3, a discrete-event network simulator, in conjunction with ns3-ai, a reinforcement learning gym, to implement and evaluate our approach (named ReinRate) under varying network scenarios, with and without interference from competing transmitters. The extensive simulations conducted under static and dynamic scenarios clearly demonstrate our method's superiority over traditional strategies like the Minstrel and Ideal rate adaptation algorithms, with up to 102.5% and 30.6% higher network throughput in network scenarios without interference, and by up to 35.1% and 66.6% in network scenarios with interference.

Index Terms—Wireless Network, 802.11, Reinforcement Learning, Rate Adaptation

I. INTRODUCTION

The continuous evolution of Wi-Fi standards, encompassing IEEE 802.11n (Wi-Fi 4), IEEE 802.11ac (Wi-Fi 5), and the recent IEEE 802.11ax (Wi-Fi 6), has heralded the introduction of multifaceted configuration parameters in both the physical (PHY) and medium access control (MAC) layers. It is very challenging to tune these parameters to achieve optimal network performance due to the dynamically changing wireless environment, varying network interference and traffic load conditions, the different capabilities of heterogeneous end-devices and the need to satisfy the Quality-of-Service (QoS) requirements from end-users. Central to the efficient operation of these networks is the Modulation and Coding Scheme (MCS) parameter, which plays a pivotal role in determining the network throughput. Rate adaptation is the mechanism used to select the optimal MCS. However the innate dynamism of the wireless channel pose considerable challenges for static rate adaptation algorithms. Traditional algorithms such as Minstrel [1] and Minstrel High Throughput (HT) [2] exhibit limitations, especially in rapidly changing environments. The need for agile, responsive, and adaptive algorithms becomes paramount in such settings.

Enter the realm of Machine Learning (ML), specifically Reinforcement Learning (RL). RL offers a potential avenue to develop algorithms that can learn and adapt to the dynamic, time-varying characteristics of wireless channels. Our research endeavors to harness the capabilities of RL to address the perennial challenge of optimal rate-adaptation in 802.11 wireless networks. In this paper, we propose an innovative rate adaptation algorithm (named ReinRate¹) based on a reinforcement learning approach. Our proposed solution uniquely employs the REINFORCE [3] reinforcement learning algorithm, which is traditionally used in areas such as game theory and robotics, to the domain of rate adaptation. Our policy network architecture provides the flexibility to adapt to changing network conditions by exploring different actions and exploiting the optimal ones based on an epsilon-greedy policy.

Traditional rate adaptation algorithms such as the Minstrel and Ideal [4] algorithms typically rely on a limited number of observations such as packet loss rate or Signal-to-Noise Ratio (SNR). In contrast, our proposed ReinRate algorithm incorporates as input a broader set of observations, including received signal strength, contention window size, current MCS, and throughput. This comprehensive view allows for a more nuanced response to varying network conditions. To ensure the algorithm's efficiency, the reward function is formulated using the difference in the current network throughput from its theoretical maximum. ReinRate is able to learn and determine the optimal MCS that minimizes the observed reward, thereby achieving maximal network throughput. Extensive simulations on the ns-3 network simulator show that our ReinRate algorithm consistently outperforms the traditional Minstrel and Ideal rate adaptation algorithms by up to 102.5% and 30.6% higher network throughput in network scenarios without interference, and by up to 35.1% and 66.6% in network scenarios with interference.

The remainder of this paper is organized as follows. Section II discusses related work on reinforcement learning approaches to rate adaptation, while Section III presents the use of the REINFORCE algorithm in our proposed algorithm. The experimental evaluation of the ReinRate algorithm is presented in Section IV, followed by the conclusion in Section V.

¹<https://github.com/lofimizumo/80211-reinrate>

II. RELATED WORK

Reinforcement Learning (RL) has carved a niche in Wi-Fi rate adaptation, emphasizing the paradigm of experiential learning to optimize wireless communications. Distinct from open-loop strategies such as Minstrel HT [2], RL-centric approaches leverage feedback loops to comprehend and act upon the ever-evolving changes in network parameters. In essence, RL methodologies immerse the algorithm within its operational environment, typically the wireless channel. The algorithm iteratively interacts with this environment, refining its strategies through a systematic trial-and-error approach to discover the optimal rate selection that optimizes a defined reward, often throughput.

In recent years, there have been several studies that employed RL in their rate adaptation algorithms. In [5], a deep reinforcement learning (DRL) based rate adaptation algorithm was proposed that considers the implications of network collisions in its reward function. The rate adaptation problem was formulated as a Markov Decision Process (MDP) and solved using the Deep Q-Network (DQN) algorithm. In [6], the authors investigated the use of DRL algorithms such as the natural actor-critic (NAC) and proximal-policy optimization (PPO) algorithms in rate adaptation to optimize network throughput. The work in [7] used the Q-learning algorithm in a reinforcement learning framework specifically tailored for the rate adaptation objective. The approach used the sender's contention window size as the observed network state and the number of acknowledged packets from receivers as the reward during discrete time intervals. In [8], the RL agent was trained based on the observed signal-to-noise-ratio (SNR) of received frames, with the product of the frame error rate and normalized MCS used as the reward function. A pivotal aspect of the approach was the use of a deep Q-learning network to train the reinforcement learning agent. In [9], the rate adaptation problem was first modeled as a 3D maze problem with the three major rate features MCS, MIMO mode, and bandwidth used as the coordinates of the 3D maze, and then solved using the Double DQN method. The majority of these works only evaluated their algorithms under scenarios with no network interference, while the works in [5] [9] only evaluated their algorithms in scenarios where both the sender and receiver nodes are being interfered with by other sender and receiver nodes. In contrast, in our work, we systematically evaluate the performance of our ReinRate algorithm under several network scenarios, with and without interference from competing senders. We also use the REINFORCE algorithm to solve the rate adaptation problem, in contrast to the use of the DQN algorithm in these works.

III. THE REINFORCE ALGORITHM AND ITS APPLICATION IN RATE ADAPTATION

The field of Reinforcement Learning (RL) offers a suite of algorithms tailored to optimize decision-making over time, catering to environments with stochastic dynamics. A seminal algorithm in this space is REINFORCE [3], which stands as

a policy gradient method aimed at maximizing the expected cumulative reward by directly optimizing the policy.

A. Foundations of REINFORCE

The REINFORCE algorithm operates under the principle that if an action taken in a particular state results in a positive outcome (or reward), the probability of choosing that action again in the future should increase. It achieves this through gradient ascent on the expected reward. Mathematically, the objective is to maximize:

$$J(\theta) = E_{\pi_{\theta}} \left[\sum_{t=0}^T r_t \right] \quad (1)$$

where π_{θ} is the policy parameterized by θ and r_t represents the reward at time t . The gradient of this objective with respect to the policy parameters is given by:

$$\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) R_t \right] \quad (2)$$

Here, R_t is the cumulative reward from time t onwards, and s_t and a_t are the state and action at time t , respectively.

B. REINFORCE in Rate Adaptation

In the context of rate adaptation for Wi-Fi networks, the REINFORCE algorithm is aptly suited to learn and determine the optimal MCS in a dynamic environment. In our application, the AP is transmitting data to the station, and hence the rate adaptation algorithm is running in the AP. The state is represented by the received signal strength (RSS) from acknowledgements (ACK) received by the AP, indicating the current wireless channel condition, in conjunction with the contention window size (CW), the current Modulation and Coding Scheme (MCS), and instantaneous throughput.

For each state, the set of possible actions corresponds to the available MCS indices. After transmitting a packet using a specific MCS (action), the AP receives an ACK, providing immediate feedback on the success or failure of the transmission. The difference in throughput from its theoretical maximum based on the chosen MCS becomes the reward signal, guiding the algorithm's learning process.

By leveraging the REINFORCE algorithm, our rate adaptation strategy iteratively updates its policy based on the observed rewards. This continuous learning process ensures that the AP consistently selects an MCS that maximizes the throughput, catering to the dynamic variations in the wireless channel. In essence, by applying REINFORCE to rate adaptation, we harness the algorithm's capacity to learn optimal strategies from raw feedback, enabling our Wi-Fi networks to achieve higher efficiencies, even in fluctuating channel conditions.

C. Implementation of ReinRate

The architecture of our algorithm is shown in Algorithm 1, consisting of a PolicyNetwork class to generate action probabilities through a softmax-activated fully connected network,

Algorithm 1: ReinRate: REINFORCE based Wi-Fi Rate Adaptation

Data: State size, Action size, Learning rate

Result: Rate Adaptation Policy

```
1 Class PolicyNetwork:
2   Initialize: Fully connected network layers (state
    size  $\rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow$  action size)
3   Forward pass: Pass input through FC layers,
    apply softmax at the end
4 Class REINFORCEAgent:
5   Compute the action probabilities using the policy
    network
6   if Training mode and random value  $< \epsilon$  then
7     Choose a random action
8   else
9     Sample an action based on the computed
    probabilities
10 Function update:
11   Initialize cumulative return  $R = 0$ 
12   Compute the returns for each time step
13   Normalize the returns
14   Compute the policy loss using the saved log
    probabilities and returns
15   Backward propagate the policy loss and update the
    policy using the optimizer
16   Clear the saved rewards and log probabilities
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and a REINFORCEAgent class employing an epsilon-greedy strategy for action selection. The update function calculates and normalizes cumulative returns at each timestep, computes policy loss, and backpropagates it to update the policy network via optimization.

Observation: In our proposed architecture, the observation or state (S_t) is characterized by several key variables obtained from the network dynamics. These include the RSS of the ACK frames at the AP, the current contention window size, the current MCS and throughput. The RSS provides a real-time assessment of the wireless channel condition at time t , denoted as RSS_t . Therefore, incorporating all these variables, the observation S_t can be formalized as:

$$S_t = f(RSS_t, CW_t, MCS_t, TP_t) \quad (3)$$

where $f(\cdot)$ is a function that amalgamates RSS, contention window size (CW_t), the current MCS (MCS_t), and throughput (TP_t) into a state representation.

Action: The action (A_t) in our framework refers to the selection of the MCS for the data packets. Using the REINFORCE algorithm, the system adaptively selects an action given the current state, formalized as:

$$A_t = g(S_t) \quad (4)$$

where $g(\cdot)$ is the policy function derived from the REINFORCE algorithm that dictates the action to take given the current state of the network.

Reward: The reward function (R_t) in our proposed system is formulated to encourage actions that lead to the maximization of the current throughput in the network. The system grants higher rewards when actions result in elevated throughput and correspondingly, assigns lower rewards when actions result in lower throughput.

In our reward function, we calculate the corresponding theoretical maximum throughput values for the eight MCS indices, based on the channel bandwidth and Wi-Fi standard (802.11n). This provides an indicator of the discrepancy between the current throughput and the theoretical maximum throughput. This deviation guides the system to optimize the MCS selection, moving closer to the optimal state. The reward function (h) can be defined to reflect this as:

$$R_{t+1} = h(A_t, S_t, S_{t+1}, T(A_t)) \quad (5)$$

where $T(A_t)$ represents the theoretical maximum throughput of the action A_t . Here, R_{t+1} is the reward at time $t+1$, which depends not only on the action taken at time t and the state transition from S_t to S_{t+1} , but also the difference between the current and theoretical maximum throughput. Thus the reward function encourages the system to adapt the transmission rate towards the theoretical maximum throughput, hence improving the overall network performance.

Simulation Environment: Simulations and evaluations in our research are conducted within the ns-3 environment, a widely recognized discrete-event network simulator. A significant part of our setup is the integration of ns3-ai [10], a reinforcement learning gym. Our rate adaptation algorithm operates within this gym. The variables required for observation, such as RSS, current contention window size, MCS, and throughput, are collected in ns-3 and then written into a shared memory chunk accessible by ns3-ai. This seamless integration of ns-3 and ns3-ai establishes a bi-directional communication, providing a real-time interface for the reinforcement learning model.

Training: The policy network in our architecture takes the current state S_t as its input and generates corresponding output actions. The subsequent MCS selection is made utilizing an ϵ -greedy policy. Under this approach, the agent predominantly selects actions that it perceives to be optimal, or 'greedy', doing so with a probability of $1 - \epsilon$. However, to ensure a degree of exploration and facilitate convergence to an optimal policy, the agent also selects actions randomly with a probability of ϵ . The procedure for collecting environment observation values and selecting the MCS unfolds as follows in our proposed architecture. After every 20 packets transmitted from the AP, the system collects and calculates the environment observation values from ns-3. These observation values are then relayed to ns3-ai, which determines the next MCS. The newly determined MCS is then applied to the forthcoming batch of packets scheduled for transmission from the AP.

TABLE I: ns-3 experiment settings

Parameter	Value
Size of Packet	1420 bytes
Simulation Time	80 s
Loss Model	MatrixPropagationLossModel, 30 dB
Transmit Power	20 dBm
Channel Width	20 MHz
Error Model	NistErrorRateModel
CCA Sensitivity	-110 dBm
Rx Noise Figure	0 dBm

TABLE II: Hyperparameters and settings for the REINFORCE algorithm

Hyperparameter/Setting	Value
State Size	10
Action Size	1
Learning Rate (lr)	1e-4
Epsilon (eps)	1e-8
Policy Network Layers	[state_size, 16, 16, 16, action_size]
Activation Functions	ReLU
Output Activation	Softmax
Initialization	Xavier Uniform
Epsilon for Action Selection	0.3
Discount Factor (gamma)	0.99

IV. EXPERIMENTAL EVALUATION

In this section, we outline the experimental framework established for the purpose of evaluating the effectiveness of our proposed rate adaptation algorithm ReinRate. The focus of our experimental evaluation lies in comparing ReinRate with existing rate adaptation algorithms, specifically the Minstrel HT [2] and the Ideal [4] algorithms. The primary metric under consideration for our evaluation is the network throughput.

A. Experiment Settings

Table I summarizes the experiment settings used in our ns-3 simulations, where the AP is transmitting traffic to the station that saturates the wireless link throughout the simulation duration. The parameters used by our REINFORCE-based rate adaptation algorithm are shown in Table II.

B. Baseline Models

The baseline models for our experiment consist of the Minstrel HT and Ideal algorithms. Minstrel HT, a commonly used rate adaptation strategy, implements a statistical, sample-based approach to rate selection. The Ideal RA algorithm, in contrast, operates by maintaining a record of the Signal-to-Noise Ratio (SNR) of every received packet at each station, and transmitting this SNR back to the sender (AP) via an out-of-band mechanism. This SNR is then utilised to select an MCS, guided by a set of SNR thresholds that are derived from a target Bit Error Ratio (BER) and MCS-specific SNR/BER curves.

C. Network Topologies

Our experimental evaluation is conducted under five distinct topology settings, intended to thoroughly evaluate the performance of our proposed algorithm in various scenarios with and

without network interference. Figure 1 shows the topological settings for these five scenarios.

- 1) **Scenario 1: Constant Moving:** The initial topology in Figure 1a shows a single Access Point (AP) and a single station (STA) positioned x meters apart. To simulate deteriorating channel conditions and evaluate the responsiveness of our algorithm to changes in channel quality, the station is steadily moved away from the AP over a period of 80 seconds. The movement speed is set at a pace of 0.5 m/s to ensure gradual changes in the channel conditions.
- 2) **Scenario 2: Random Walk:** The second topology in Figure 1b makes use of a Random Walk mobility model, simulating a more dynamic and unpredictable wireless network scenario. In this model, the station moves randomly within a pre-defined linear line, introducing significant variations in channel conditions. This scenario serves as a rigorous test for our rate adaptation algorithm's resilience and adaptability. The mobility model used is `ns3::randomwalk2dmobilitymodel`, with the range of the linear line between 0 to 24 meters away from the AP. The speed of the station is maintained at a constant 3.0 m/s.
- 3) **Scenario 3: 1 AP, 1 STA, 1 Interfering AP:** In this configuration in Figure 1c, we deploy a single AP with a single station positioned 10 meters away from the AP. The Interfering AP (IAP) progressively moves away from the station, starting at 8m and extending to 11m. This basic configuration serves as a benchmark to assess the performance of the rate adaptation algorithm in the presence of network interference.
- 4) **Scenario 4: 1 AP, 1 STA, 1 Interfering AP with Random Movement:** In Figure 1d, the scenario employs a configuration similar to Scenario 3. However, the distinguishing feature here is that the IAP is set to move in a random walk pattern within a 30m x 30m rectangle.
- 5) **Scenario 5: 1 AP, 1 STA, 3 Interfering APs, Static Scenario:** In this setup shown in Figure 1e, we simulate a high-interference environment with four APs transmitting packets to a single station (STA). The goal is to study the response of the rate adaptation mechanism to increased interference levels. The AP, three Interfering Access Points (IAPs), and STA are arranged in a star formation, with the STA centralized. The variable x denotes the distance between the IAPs and the STA, evaluated for values ranging from 8.75m to 10m.

We conduct 20 simulation runs for each network scenario and throughput results are averaged over these runs. The five distinct network scenarios, incorporating both static and dynamic topologies, with and without network interference, enable a comprehensive evaluation of our proposed algorithm under a range of network conditions.

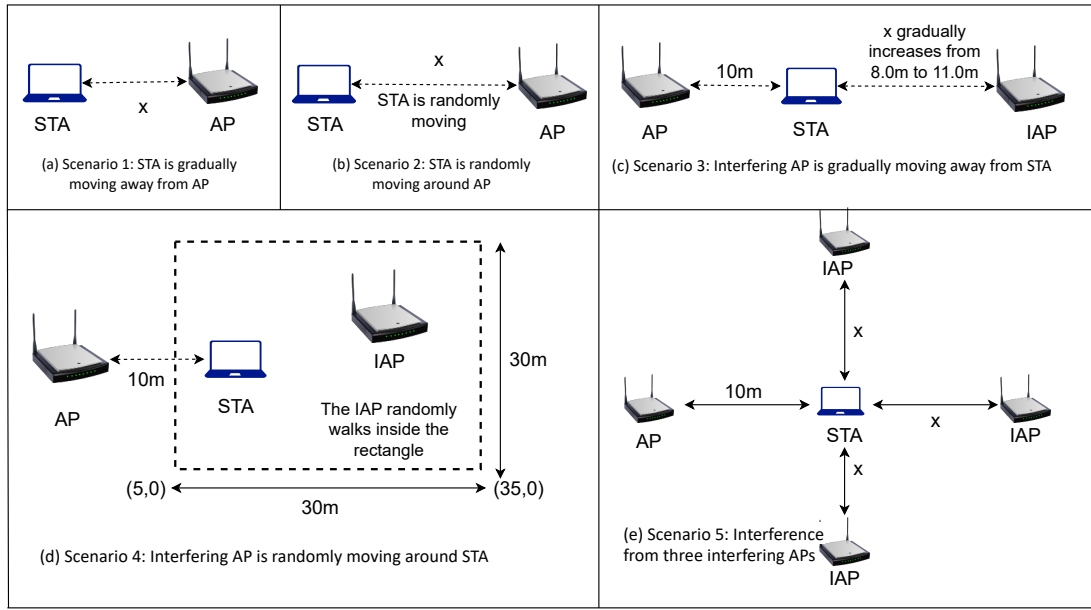


Fig. 1: Topologies of five network scenarios

D. Discussion of Experimental Results

1) *Scenario 1: Constant Moving*: Figure 2a shows that the throughput performance decreases as the STA gradually moves away from the AP. We can see the distinct step declines in the throughput, corresponding to the different MCS levels chosen by the rate adaptation algorithms as the channel quality gradually degrades. Overall we can see that our proposed rate adaptation algorithm ReinRate tracks the throughput performance of the Ideal algorithm, and clearly outperforms the throughput performance of the Minstrel algorithm. ReinRate is shown to be better at adapting its performance in the presence of mobility-induced channel fluctuations, compared to Minstrel.

2) *Scenario 2: Random Walk*: Figure 2b shows that all three rate adaptation algorithms exhibit fluctuations in their instantaneous throughput values as the STA randomly moves around the AP. However, Minstrel demonstrated notably poorer overall throughput performance and greater throughput fluctuations compared to both the ReinRate and Ideal algorithms. ReinRate, in contrast, produced a better throughput performance, suggesting it is more capable to quickly adapt to rapid changes in channel conditions.

3) *Scenario 3: One Interfering AP Moving Away from STA*: In this setup, we positioned one AP and one STA 10 meters apart. This relatively short distance² was determined from the results in Scenario 1, marking the point where the throughput in all three algorithms begin to drop from the maximum throughput. We also have an Interfering AP (IAP) gradually moving further away from the STA as we measure the throughput of the AP's transmission to the STA. Figure 2c shows that at a distance $x = 8\text{m}$, the strong interference from

the IAP caused all of the AP's transmissions to be dropped at the STA. As the IAP gradually moves away until $x = 11\text{m}$ (therefore causing a reduction in its interference), we see a sharp increase in the throughput until it reaches the maximum throughput, indicating a swift recovery of the AP's throughput. Throughout this range, ReinRate consistently outperformed the other two rate adaptation algorithms, indicating its ability to optimally select the MCS value in the face of network interference.

4) *Scenario 4: One Interfering AP Randomly Moving Inside a Rectangle*: This scenario is similar to Scenario 3, with the main difference being the Interfering AP is moving randomly within a 30m x 30m rectangular area. In this scenario, we want to evaluate the performance of the rate adaptation algorithms in dynamic network interference conditions. The results in Figure 3 shows the AP's average throughput derived from 20 simulation runs. We see that ReinRate achieves the highest network throughput compared to the other two rate adaptation algorithms, indicating its superior performance in the face of dynamic network interference.

5) *Scenario 5: 1 AP, 1 STA and 3 Interfering APs*: In this scenario, three IAPs are symmetrically positioned at an equal distance x from the centrally located STA. We measure the AP's throughput in the face of higher interference from multiple IAPs. Figure 4 clearly shows that ReinRate maintains a superior network throughput performance in comparison to the other two rate adaptation algorithms across the various interference distances³. This indicates that compared to the other two algorithms, ReinRate is better able to optimally select the MCS value that results in higher throughput in conditions characterized by high network interference.

²In our simulations, the wireless channel is a lossy channel.

³The effective interference range in our simulations is relatively short, similar to that seen in Scenario 3's results.

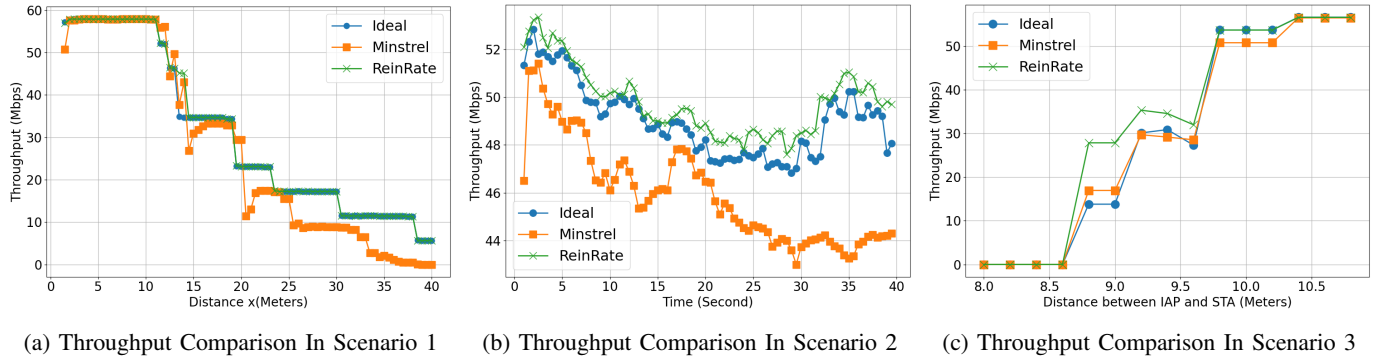


Fig. 2: Scenario 1-3: Throughput Comparison Across Three Scenarios

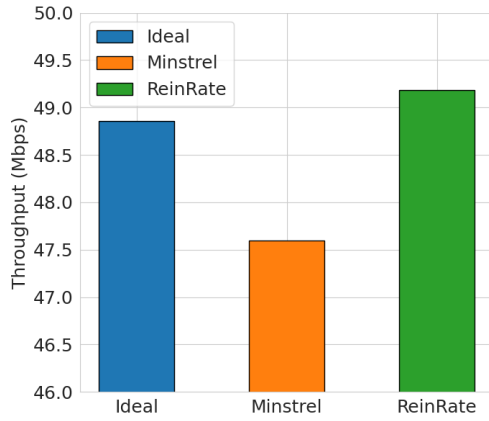


Fig. 3: Throughput Comparison in Scenario 4

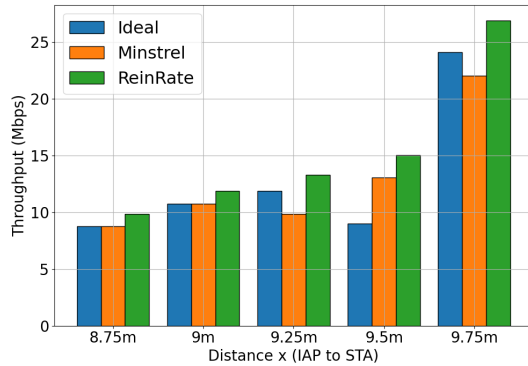


Fig. 4: Throughput Comparison in Scenario 5

V. CONCLUSION

In this paper, we have introduced a novel rate adaptation strategy for 802.11 wireless networks, named ReinRate, that is based on the REINFORCE reinforcement learning algorithm. ReinRate is designed to optimize the instantaneous network throughput, positioning it as a more adaptive and responsive solution in diverse network conditions. ReinRate's adaptability comes from its capability to synthesize a broader range of network observations, incorporating metrics like the received

signal strength, contention window size, the MCS and throughput. This comprehensive perspective ensures ReinRate's aptitude for finely-tuned responses to network variations. With the REINFORCE algorithm at its core, ReinRate can iteratively discern the best rate adaptation actions corresponding to different network states. We have compared ReinRate's performance with the traditional Minstrel and Ideal algorithms in diverse network scenarios, with and without network interference. Experimental results show that ReinRate consistently outperforms both algorithms, by up to 102.5% and 30.6% in network scenarios without interference, and by up to 35.1% and 66.6% in network scenarios with interference. Our results also show that ReinRate is able to quickly adapt to dynamic changes in channel conditions, compared to the two traditional algorithms.

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